Neelesh Kumar Tripathi

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Education

The National Institute of Engineering — **Mysore, Karnataka**
Bachelor of Engineering in Information Science — **CGPA: 8.60/10** — *Sep. 2023 – May 2027*

Little Flower School — **Gorakhpur, Uttar Pradesh**
Senior Secondary Education (Class XII) — **Percentage: 91%** — *July 2022*

Projects

Crackd – AI Powered Placement Assistance Platform (Demo) — **May 2026**
MERN Stack, MongoDB, JWT, RAG, Gemini API, RESTful API, Axios, Redis, CI/CD Pipeline — *Repository Link*

- Built a full-stack placement platform with **Node.js**, **Express.js**, and **MongoDB** where students can browse company-wise interview experiences, get personalized prep plans, and track their roadmap progress in one place.
- Integrated the **Google Gemini API** to generate personalized, role-specific study roadmaps with weekly plans and topic-wise checklists, tailored to each student's available preparation time, current strengths, and weak areas.
- Developed a **RAG-based interview assistant** using **Gemini embeddings** and **cosine similarity search** to retrieve relevant interview experiences from **MongoDB** and answer student questions with accurate, grounded responses.
- Built secure **RESTful APIs** with **JWT authentication**, password hashing, and college-email domain validation.
- Added **Redis caching** for frequently accessed company data, cutting repeat queries with a graceful cache fallback.
- Set up **CI/CD with GitHub Actions** to automate linting, builds, tests (**Jest**, **Supertest**), and deployment.

LLM Cost & Latency Optimizer Gateway — **October 2026**
Python, FastAPI, Sentence-Transformers, NumPy, SQLite, Pydantic, Groq, Gemini, OpenRouter — *Repository Link*

- Built a **Python/FastAPI** gateway that routes prompts to cheap or powerful LLM tiers by task type and length.
- Implemented a **semantic cache** using **Sentence-Transformers** (all-MiniLM-L6-v2) embeddings and **NumPy** cosine similarity at a 0.85 threshold, reusing answers for repeated and paraphrased prompts without extra LLM calls.
- Integrated **Groq**, **Gemini**, and **OpenRouter** through **OpenAI SDK**-compatible endpoints with ordered multi-provider fallback, returning HTTP 503 instead of a fabricated answer when every configured model fails.
- Tracked tokens, latency, and cost in **SQLite**, with a **/stats** endpoint reporting cache hit rate and savings.

FinNIE – Campus Lost & Found Platform — **September 2024**
Flask, MySQL, HTML, CSS, Jinja2 — *Repository Link*

- Developed a full-stack web application enabling users to report, browse, and manage lost and found items.
- Implemented a **Flask** backend with **MySQL** for data storage, recording $\sim$**10 ms** average database query latency.
- Engineered keyword-based search and category filtering to significantly improve item lookup efficiency.
- Implemented **API rate limiting** using **Flask-Limiter** to prevent abuse and improve backend security.

Technical Skills

Languages & Libraries: C, C++, Python, JavaScript, SQL, NumPy, Sentence-Transformers, Pydantic
AI/LLM: Gemini API, Groq, OpenRouter, RAG, Embeddings, Semantic Caching, LLM Routing, Prompt Engineering
Backend Development: Node.js, Express.js, FastAPI, Flask, OpenAI SDK, RESTful APIs, JWT Authentication
Databases & Caching: MongoDB, MySQL, SQLite, Redis
Developer Tools: Git, GitHub, GitHub Actions, Jest, Supertest, Postman, VS Code, Render, Vercel
Core Concepts: Data Structures & Algorithms, OOP, Computer Networks, Operating Systems

Achievements

Problem Solving & Hackathons — **Codolio Profile**

- Solved **500+ Data Structures and Algorithms** problems on LeetCode, GeeksforGeeks, and other coding platforms.
- Ranked **4th among 100+ teams** at DevJams, VIT, building an AI Sign-to-Text and Text-to-Sign platform in 24 hours.

Extra-Curriculars & Leadership

The BYTE Club, NIE — **August 2024 – January 2026**
Management Core Member — *The National Institute of Engineering, Mysuru*

- Led the intercollegiate flagship tech event **HackForge 2025** as **Event Lead**, coordinating teams across the state.
- Organized **10+ workshops and coding events** for **200+ students**, driving hands-on learning and collaboration.
- Coordinated logistics, sponsorships, and strategic planning to ensure seamless club operations and event delivery.